Mohaddeseh (Sahar) Hosseinzadeh

AI research engineer with a focus on healthcare applications, biomedical signal analysis, and neuroscience, particularly interested on developing computational tools for clinical applications. I apply deep learning to large-scale neural and cardiac data, including fMRI, EEG, ECG, EMG, and LFP and I work across the complete delivery cycle: data auditing and preprocessing, architecture selection, model training and evaluation, and containerized deployment. Currently, I am developing ECG models for myocardial infarction detection at the Barcelona Supercomputing Center.

PORTFOLIO : Synapticsahar.github.io Github LinkedIn

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Interests: Biomedical Signal Analysis | Digital health | Cardiovascular AI | NeuroAI | Computational Neuroscience | Computational Cardiology | Deep Learning

EXPERIENCE

AI Research Engineer Intern – ECG Foundation Model for STEMI/NSTEMI detection from 12-Lead ECG (Cardiovascular AI)
Barcelona Supercomputing Center, Digital Health Unit — *05/2026 – present*

- Built the STEMI and NSTEMI detection pipeline on a private Hospital Sant Pau cohort of 10,491 12-lead ECG recordings, from data audit and preprocessing through training and evaluation on BSC HPC.
- Benchmarked ECGFounder against machine learning models and CNN-based architectures trained from scratch under a shared evaluation protocol, scored on AUROC, AUPRC and sensitivity at fixed specificity with subgroup breakdowns.
- Selected the model on licensing and commercial-use terms alongside performance, so the hospital can deploy it and develop it further rather than keep it restricted to research use.
- Delivered a documented, reproducible pipeline ready for clinical development, with results planned for publication.

Master's Thesis Researcher – Brain-State Dynamics in Vortex Space Using fMRI
Supervisors: Gustavo Deco and Rubén Moreno-Bote, UPF — *12/2025 – 07/2026*

- Built a seven-state brain-state model from 7T resting-state fMRI (182 Human Connectome Project participants), using local Kuramoto order parameter, PCA and k-means.
- Linked transition and occupancy entropy to cognition under permutation and bootstrap testing, positive in 26 of 28 tests; ridge-dominant elastic net on 49 transition features reached cross-validated $r = 0.24$–0.29, beating SVR, random forests and gradient boosting.
- Designed a novel exponential distance model of state transitions (group matrix at $r = 0.90$, $r = 0.54$ under leave-one-participant-out CV) and evaluated recursive Maximum Occupancy Principle models testing whether transitions favour states that preserve access to more future states. Thesis PDF | GitHub Repository | Presentation Slides

Research Collaborator – Sharp Wave Ripple Detection from Hippocampal LFP
CMC Lab, TU Dresden — *03/2025 – 06/2026*

- Built a cross-session ripple-detection pipeline using STFT and non-negative matrix factorization to extract spectral features from hippocampal LFP.
- Reproduced the rippl-AI reference models on raw LFP as controlled baselines and benchmarked dense Informer, CNN1D, InceptionTime, and temporal convolutional networks on factorization-derived features using identical data splits and event-level evaluation criteria; all feature-based models outperformed the strongest reproduced raw-LFP baseline.

AI Research Engineer – EEG-Based Affective State Decoding
IKHC, Clinical Lab, Tehran — *01/2025 – 07/2025*

- Built an EEG emotion decoder on DEAP to read shopper responses to product presentations, paired with eye-tracking glasses, with a full MNE-Python pipeline (bandpass filtering, ICA artifact removal, epoching, band-power features).
- Trained EEGNet for four-class valence-arousal classification under leave-one-participant-out, reaching 58.0% mean accuracy vs a 35.8% majority baseline, with macro $F_1 = 0.54$ across 32 folds.
- Served the model in real time via FastAPI, Streamlit and Docker on AWS EC2.

AI Research Engineer – Deep Learning on EEG and EMG for Wearable BCI
Myant Inc., Canada — *03/2024 - 09/2024*

- Built the preprocessing stack for EEG and EMG streamed live from garments, targeting the two failure modes specific to textile electrodes: movement artifacts, and slow drift in electrode contact impedance that a fixed filter setting cannot follow.
- Developed an eye-movement tracking model using headband signals that achieved 91.6% accuracy, and a recurrent neural network that mapped forearm EMG from glove hardware to continuous movement commands for AR/VR control with $R^2 = 0.75$.

Full Stack Software Developer
Rahnema College — *06/2021 – 08/2022*

- Built and deployed a full-stack web app end-to-end.

EDUCATION

M.Sc., Computational Neuroscience
Brain and Cognition, Pompeu Fabra University, Barcelona (GPA 8.38/10) — *2025 – 2026*

B.Sc., Industrial Engineering
Sharif University of Technology, Tehran — *2017 – 2023*

Bachelor's Thesis – Deep Q-Network for the Beer Game

- Built a multi-agent reinforcement learning pipeline using DQN with reward shaping to train agents that act only on local information and minimize a shared global cost.
- Teaching Assistant, Linear Algebra (40+ undergraduates).

SELECTED PROJECTS

Agentic ECG Decision Support
- Fine-tuned ECGFounder on PTB-XL for five diagnostic superclasses and integrated the calibrated classifier as an LLM tool with confidence-based abstention.
- Built an agentic clinical reasoning system combining ECG classification, HL7 FHIR extraction, guideline retrieval and schema-validated reports with citations and missing-information flags.
- Benchmarked five RAG and agent configurations using retrieval recall, citation faithfulness, hallucination rate, unsupported claims, abstention accuracy, latency and cost, with CI regression gates.

Seizure Detection with PatchTST
- Fine-tuned PatchTST on 969 hours of CHB-MIT EEG for seizure detection, evaluating sensitivity and false alarms per hour to address severe class imbalance.
- Benchmarked the model against clinical baselines and deployed it on AWS SageMaker with a FastAPI backend and Streamlit frontend.

EEG-to-Image Decoding with CLIP
- Trained a Transformer encoder on THINGS-EEG2 to align EEG segments with CLIP image embeddings using contrastive learning.
- Retrieved the three most likely image categories from EEG and used GPT-4o to generate descriptions of the viewed images.

Synaptic Plasticity Modeling in Mouse V1
- Built reward-modulated spiking neural networks in Brian2 to model VIP, SST and excitatory-cell interactions in visual learning.
- Used Allen Brain Observatory two-photon recordings from mouse V1 to compare neural activity during rewarded and passive viewing.

3D U-Net for Brain Tumor Segmentation
- Trained a 3D U-Net for volumetric brain tumor segmentation on BraTS2020, using Dice loss, data augmentation, and learning-rate scheduling.

TECHNICAL SKILLS

- **Biosignals and neuroscience tools:** ECG, EEG, EMG, LFP, fMRI, DEAP, PTB-XL, CHB-MIT EEG, THINGS-EEG2, HCP, Brats2020, Allen Brain Observatory, MNE-Python, Brian2, AllenSDK
- **Scientific Programming:** Python, MATLAB, C/C++, R, Shell
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn; Conventional Machine Learning, CNNs and U-Nets, RNNs and GRUs, Transformers, Time-Series Deep Learning, Spiking Neural Networks, Reinforcement Learning, Contrastive Learning, ECG Foundation Models, Transfer Learning, PEFT and LoRA
- **LLMs:** HuggingFace, LangChain, RAG, OpenAI API, LLM tracing
- **Computing:** AWS (EC2, SageMaker, Lambda, Lightsail), LBSC HPC cluste, Linux
- **Software Engineering:** Java Spring, Javascript, React, HTML, CSS, Git, REST APIs, FastAPI, Streamlit, CI/CD, Docker, Software Design Principles

REFERENCES

Prof. Gustavo Deco, MSc thesis supervisor — CNS Lab, UPF

Dr. Paula Petrone, RAPID-AIM project lead — Barcelona Supercomputing Center

Prof. Rubén Moreno-Bote, MSc thesis co-supervisor — TCN Lab, UPF